UEE002: PRINCIPLES OF ELECTRICAL ENGINEERING

L	T	P	Cr
3	0	2	4.0

Course Objectives: To introduce concepts of DC and AC circuits and electromagnetism. To make the students understand the concepts and working of single-phase transformers, DC motor and generators.

Introduction: Fundamental linear passive and active elements to their functional current-voltage relation, voltage source and current sources, ideal and practical sources, Kirchhoff-s laws and applications to network solutions using mesh and nodal analysis, Concept of work, power, energy, and conversion of energy.

Basic network: Current-voltage relations of the electric network by mathematical equations to analyze the network (Thevenin‟s theorem, Norton's Theorem, Maximum Power Transfer theorem) Simplifications of networks using series-parallel, Star/Delta transformation. Superposition theorem.

Concept of AC: AC waveform definitions, form factor, peak factor, phasor representation in polar and rectangular form, concept of impedance, admittance, complex power, power factor, single phase and three phase concept.

Electrostatics and Electro-Mechanics: Electrostatic field, electric field strength, concept of permittivity in dielectrics, energy stored in capacitors, charging and discharging of capacitors. ElectroMagnetism, magnetic field and Faraday's law. Magnetic materials and B-H curve. Self and mutual inductance, Ampere's law, Study of R-L, R-C, RLC series circuit, R-L-C parallel circuit. Electromechanical energy conversion.

Measurements and Sensors: Measuring devices/sensors and transducers (Piezoelectric and thermo-couple) related to electrical signals, Elementary methods for the measurement of electrical quantities in DC and AC systems (Current & Single-phase power). Concept of indicating and integrating instruments.

Practical considerations: Electrical Wiring types and accessories, Illumination system, Basic layout of the distribution system, Types of earthing, Safety devices & systems. Battery principles and types.

Laboratory

1. Familiarization of electrical circuits: sources, measuring devices and transducers
2. Determination of resistance temperature coefficient
3. Verification of Network Theorem (Superposition, Thevenin, Norton, Maximum Power
4. Transfer theorem)
5. Simulation of R-L-C series circuits for XL>XC , XL< XC
6. Simulation of Time response of RC circuit
7. Demonstration of measurement of electrical quantities in DC and AC systems.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

On the completion of course, students will be able to:

1. To apply basic concepts, network laws and theorems to solve DC circuits.
2. Signify AC quantities through phasor and compute single-phase series and parallel AC system behavior during steady state.
3. Elucidate the need of three phase system, calculations and power measurement in three-phase system.
4. Comprehend and apply the basic concepts of Electro-statics and Electro-mechanics.
5. Elucidate the principles of various measurement systems.
6. Elucidate the practical considerations of wiring systems.

Text Books:

1. Electric Machinery, (Sixth Edition) A. E. Fitzgerald, Kingsely Jr Charles, D. Umans Stephen, Tata McGraw Hill, 2003.
2. A Textbook of Electrical Technology, (vol. I), B. L. Theraja, Chand and Company Ltd., New Delhi, 2002.
3. Basic Electrical Engineering, V. K. Mehta, S. Chand and Company Ltd., New Delhi, 2012.
4. Theory and problems of Basic Electrical Engineering, (Second Edition), J. Nagrath and Kothari, Prentice Hall of India Pvt. Ltd, 1998.

Reference Books:

1. Basic Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2011.
2. Introduction to Electrodynamics, D. J. Griffiths, (Fourth Edition), Cambridge University Press, 2005.
3. Engineering Circuit Analysis, William H. Hayt& Jack E. Kemmerly, McGraw-Hill Book Company Inc, 1978.
4. Fundamentals of Electrical and Electronics Engineering, Smarajit Ghosh, Prentice Hall (India) Pvt. Ltd, 2007.